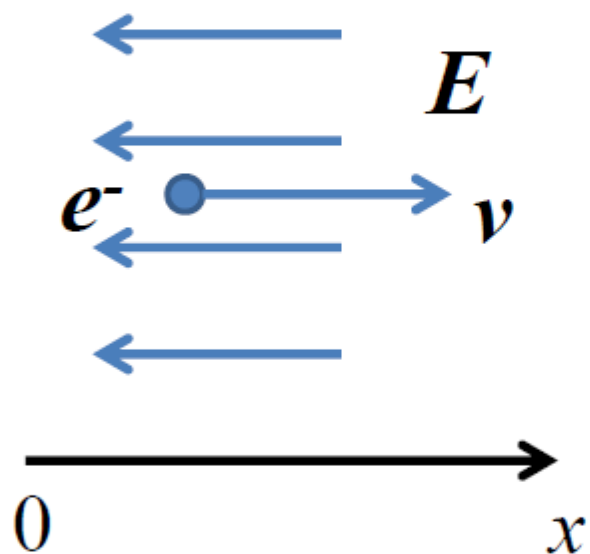


# Devices and Circuits I

Conduction in Semiconductors

# Electrons in Motion

Constant electric field in vacuum



Force on the electron:

$$F = -qE$$

Newton's 2<sup>nd</sup> Law:

$$a = \frac{F}{m} = -\frac{qE}{m}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$
$$m = 9.11 \times 10^{-31} \text{ kg}$$

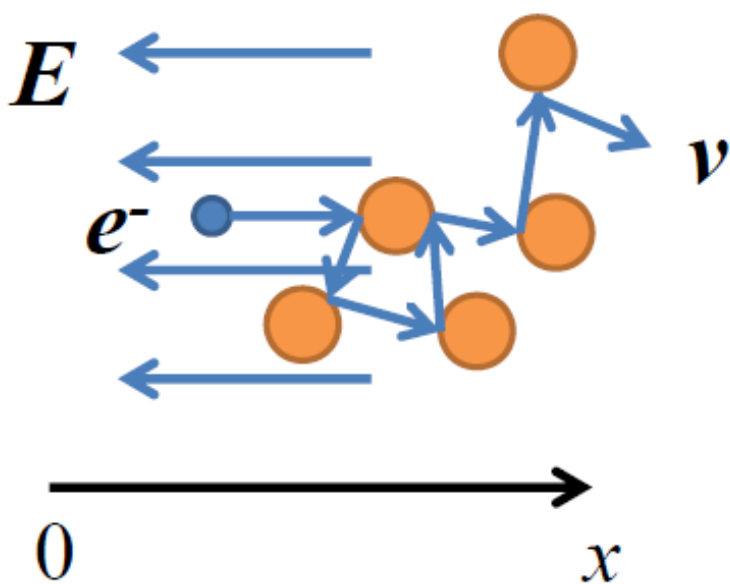
Velocity:

$$v = at \propto t$$

In vacuum, constant  $E$  field causes electrons to accelerate at **a linearly increasing** velocity

# Electrons in Motion

## Constant electric field in a solid

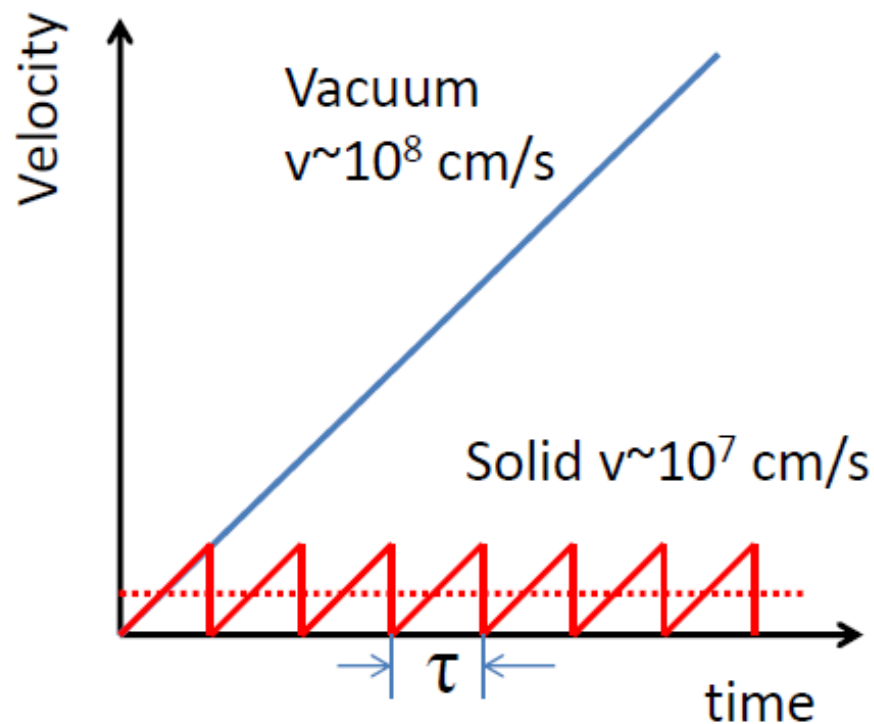


Electron flies for short intervals ( $\sim$ ps) before bumping into **scattering objects**

Result: Electron's average velocity proportional to field

$$v \propto E$$

# Electron in Motion



Electron's average velocity  
proportional to field

$$v = -\mu E$$

$\mu$ : Mobility

$$[\mu] = \frac{[v]}{[E]} = \frac{a\tau}{[E]} = \frac{q\tau}{m}$$

$$[\mu] \sim \frac{\text{cm/s}}{\text{V/cm}} = \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$$

Example:

$$|E| = 10^4 \text{ V/cm}, \mu = 200 \text{ cm}^2/(\text{V} \cdot \text{s})$$

$$|v| = \mu |E| = 2 \times 10^6 \text{ cm/s}$$

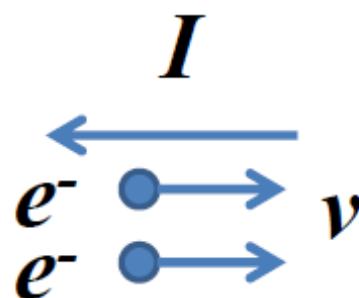
# Current and Current density

---

- Current: flow of electrons in a medium
- Direction convention: **opposite** to electron flow
- Current = charge / time

$$I = \frac{Q}{t} \quad [\text{A}]$$

– Ampere (A) = Coulomb (C) / second (s)

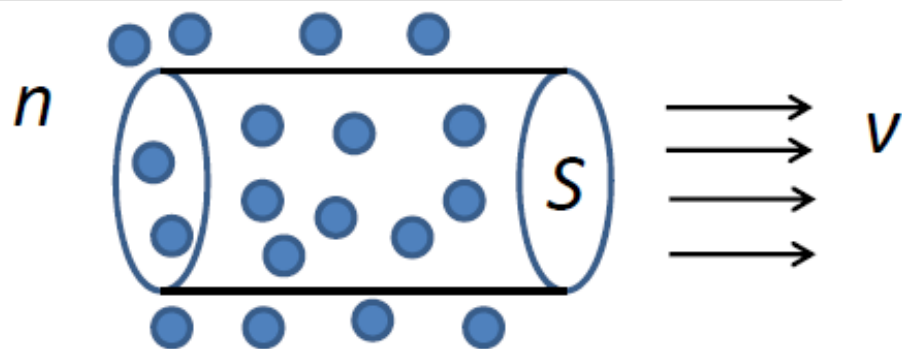


- Current density: current per unit area

$$j = \frac{I}{S} \quad [\text{A}/\text{cm}^2]$$

---

# Current density



Imagine a group of electrons are passing through a wire with speed  $v$ . Pick a small area  $S$  perpendicular to the flow direction.

During a short time period  $t$ :

- Which electrons can pass through area  $S$ ? All the electrons in the cylinder with area  $S$  and length of  $vt$
- How many of them are there?  $N_e = n \cdot V = n \cdot vt \cdot S$
- How many charges do they carry?  $Q = (-q)N_e = -qnvts$
- What is the current?  $I = Q/t = -qnvS$
- What is the current density?  $j = I/S = -qnv$

# Current Density & Conductivity

---

Current density = product of **carrier charge**, **carrier density**, and **velocity**  
( = product of **charge density** and **velocity** )

$$j = -q n v \rightarrow \text{Electron velocity [cm/s]}$$

Charge on each Electron [C]  $\leftarrow$   $\rightarrow$  Electron density [cm<sup>-3</sup>]

Intrinsic Ohm's Law: (**Drift** current density in an electric field)

$$j = -qnv = -qn(-\mu E) = qn\mu E = \sigma E$$

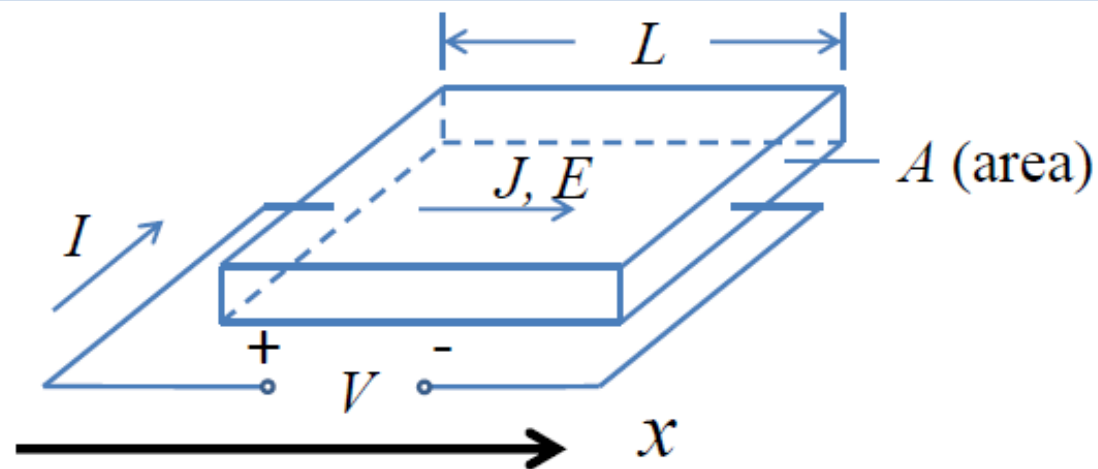
where  $\sigma = qn\mu$  is the **conductivity**.

$$[\sigma] = \frac{[j]}{[E]} = \frac{\text{A/cm}^2}{\text{V/cm}} = \frac{1}{\Omega \cdot \text{cm}}$$

**Resistivity:**  $\rho = \frac{1}{\sigma} [\Omega \cdot \text{cm}]$



# Resistance & Conductance



$$E = \frac{V}{L}$$
$$j = \frac{I}{A}$$

$$j = \sigma E \rightarrow \frac{I}{A} = \frac{1}{\rho} \frac{V}{L}$$

Extrinsic Ohm's Law

$$V = \rho \frac{L}{A} I = RI$$

where  $R = \rho \frac{L}{A}$  is the **resistance**  $[\Omega]$ .

**Conductance:**  $G = \frac{1}{R} = \sigma \frac{A}{L} [\Omega^{-1}]$ .



# Electronic Materials

---

- Electrical characterization of materials

- Insulators  $\rho > 10^5 \Omega \cdot \text{cm}$

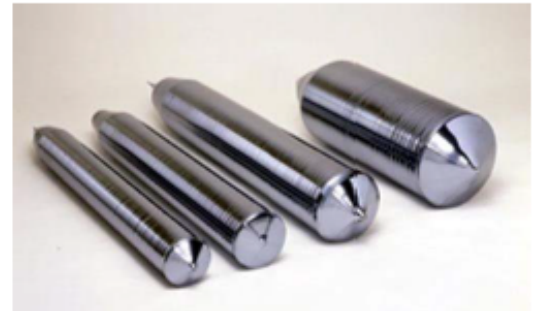
- e.g. **Diamond**  $\rho = 10^{16} \Omega \cdot \text{cm}$

- Conductors  $\rho < 10^{-3} \Omega \cdot \text{cm}$

- e.g. **Copper**  $\rho = 3 \times 10^{-6} \Omega \cdot \text{cm}$

- Semiconductors  $\rho$  in between

- Elemental semiconductors (e.g. Si)
    - Compound semiconductors (e.g. GaAs)

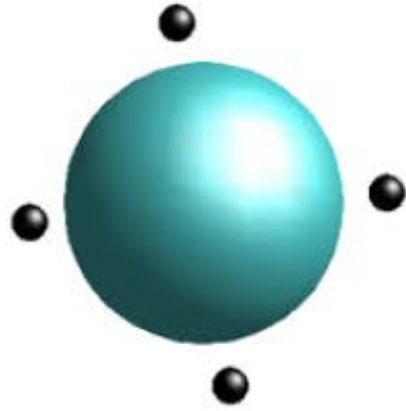


# Semiconductor Materials

|     |                                        |                                       |                                         |                                       |
|-----|----------------------------------------|---------------------------------------|-----------------------------------------|---------------------------------------|
|     | IIIA                                   | IVA                                   | VA                                      | VIA                                   |
|     | 5<br>10.811<br><b>B</b><br>Boron       | 6<br>12.01115<br><b>C</b><br>Carbon   | 7<br>14.0067<br><b>N</b><br>Nitrogen    | 8<br>15.9994<br><b>O</b><br>Oxygen    |
|     | 13<br>26.9815<br><b>Al</b><br>Aluminum | 14<br>28.086<br><b>Si</b><br>Silicon  | 15<br>30.9738<br><b>P</b><br>Phosphorus | 16<br>32.064<br><b>S</b><br>Sulfur    |
| IIB | 30<br>65.37<br><b>Zn</b><br>Zinc       | 31<br>69.72<br><b>Ga</b><br>Gallium   | 32<br>72.59<br><b>Ge</b><br>Germanium   | 33<br>74.922<br><b>As</b><br>Arsenic  |
|     | 48<br>112.40<br><b>Cd</b><br>Cadmium   | 49<br>114.82<br><b>In</b><br>Indium   | 50<br>118.69<br><b>Sn</b><br>Tin        | 51<br>121.75<br><b>Sb</b><br>Antimony |
|     | 80<br>200.59<br><b>Hg</b><br>Mercury   | 81<br>204.37<br><b>Tl</b><br>Thallium | 82<br>207.19<br><b>Pb</b><br>Lead       | 83<br>208.980<br><b>Bi</b><br>Bismuth |
|     |                                        |                                       |                                         | 84<br>(210)<br><b>Po</b><br>Polonium  |

| Semiconductor | Bandgap |
|---------------|---------|
| C (diamond)   | 5.47    |
| Si            | 1.12    |
| Ge            | 0.66    |
| Sn            | 0.082   |
| GaAs          | 1.42    |
| GaN           | 3.49    |
| InP           | 1.35    |
| BN            | 7.50    |
| SiC           | 3.26    |
| CdSe          | 1.70    |

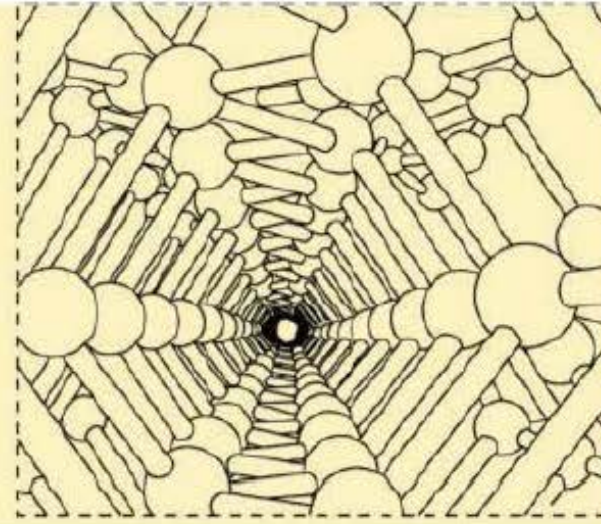
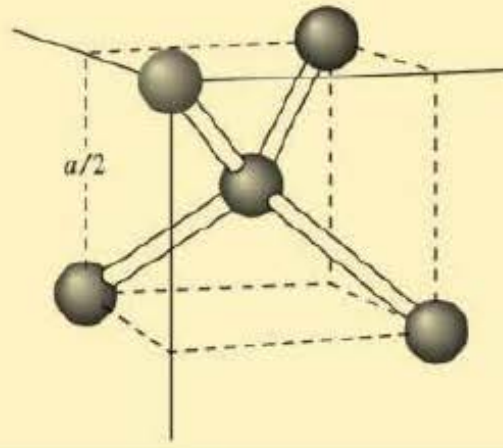
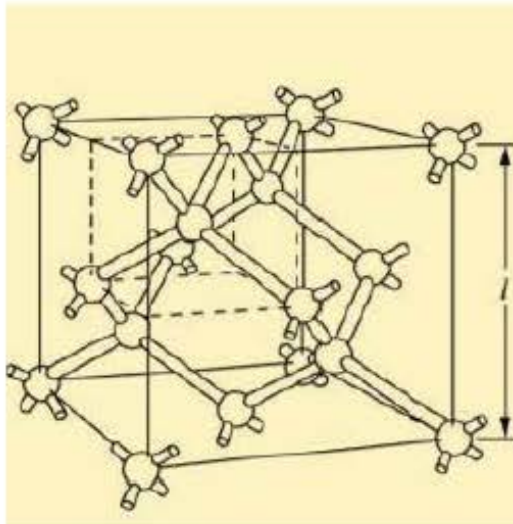
# Silicon



$^{14}\text{Si}: 1s^2 2s^2 2p^6 3s^2 3p^2$

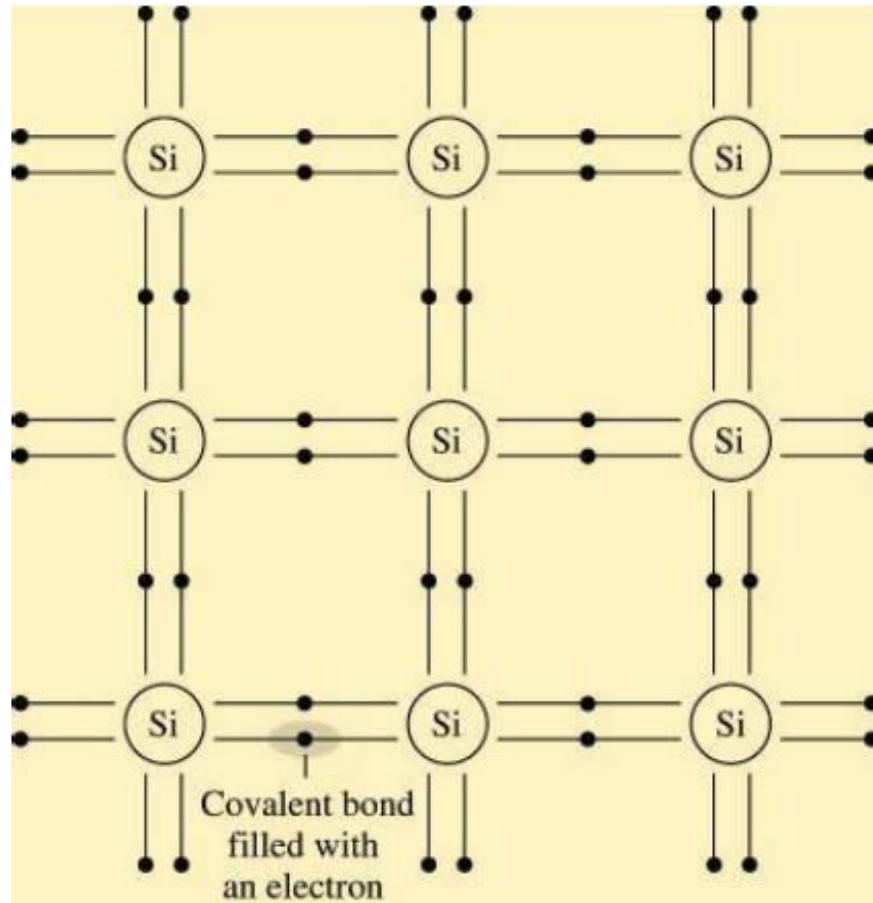
4 outer electrons ( $sp^3$  hybridization)

=> tetrahedral bonding network



# Carrier Concentration - Silicon

2d representation of Si crystal structure (  $T = 0 \text{ K}$  )



- Si forms 4 symmetric bonds
- Each bond has 2 electrons
- At  $0 \text{ K}$ , all electrons are bound by the Si atoms and are **immobile**  
 $\Rightarrow$  No **free** charge carriers for conduction  $\Rightarrow$  perfect insulator ( $\sigma = 0, \rho = \infty$ )

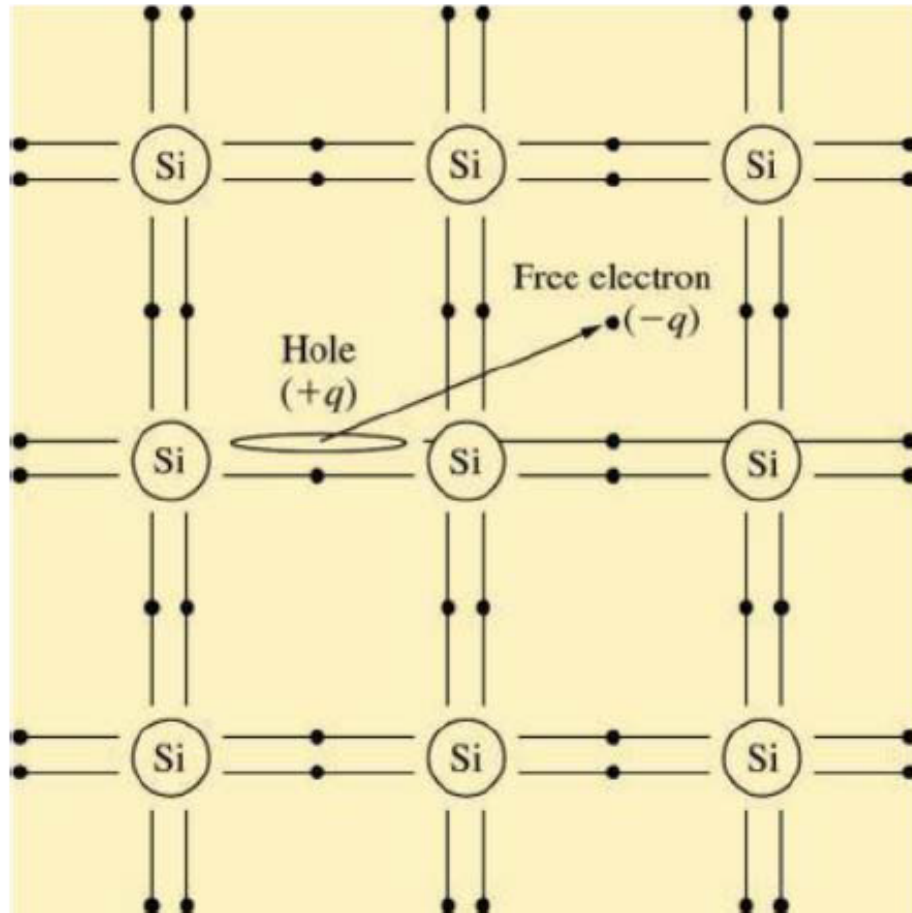
$$\sigma = q\mu n, n = 0$$



# Carrier Concentration - Silicon

---

2d representation of Si crystal structure (  $T > 0$  K)



- At  $T > 0$  K, thermal energy inside the crystal can excite small amount of bound electrons into **free electrons**, leaving a **hole** in the bond
- Density of these free electrons (and also holes) is called the **intrinsic carrier concentration**

# Intrinsic Carrier Concentration

---

- Intrinsic carrier concentration only depends on **temperature** ( $T$ ) and **material** ( $B$ ,  $E_G$ )

$$n_i^2 = BT^3 \exp\left(-\frac{E_G}{kT}\right) \text{ cm}^{-6}$$

- $E_G$  : band gap energy of the **material**, in eV.  $E_G$  is the minimum energy to break the covalent bond and excite a free electron.
- $k$  : Boltzmann's constant,  $8.62 \times 10^{-5} \text{ eV / K}$
- $T$  : Absolute **temperature** in K (Kelvin)
- $B$  : **material**-dependent parameter

$$\begin{aligned} 1 \text{ eV} &= 1.6 \times 10^{-19} \text{ J} \\ \text{K} &= ^\circ\text{C} + 273.15 \end{aligned}$$

# Intrinsic Carrier Concentration

---

- Example: Calculate intrinsic carrier concentration of Si at 300 K (RT) and 1200 K .

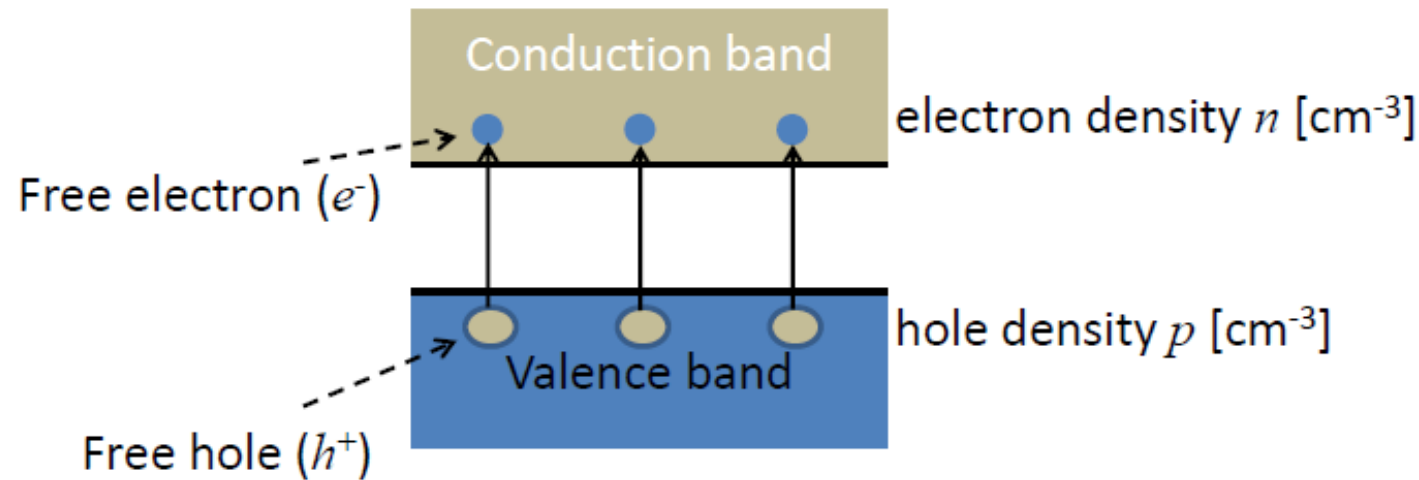
$$n_i^2 = BT^3 \exp\left(-\frac{E_G}{kT}\right) \text{ cm}^{-6}$$

- For Si:  $E_G = 1.12 \text{ eV}$ ,  $B_{\text{Si}} = 1.08 \times 10^{31} \text{ K}^{-3} \text{ cm}^{-6}$
- At 300 K,  $n_i = 6.73 \times 10^9 \text{ cm}^{-3}$
- At 1200 K,  $n_i = 6.09 \times 10^{17} \text{ cm}^{-3}$
- Silicon valence electron density:  $\sim 10^{23} \text{ cm}^{-3}$
- Fraction of bonds broken to excite a free electron
  - $\sim 10^{-14}$  for  $T = 300 \text{ K}$
  - $\sim 10^{-6}$  for  $T = 1200 \text{ K}$

8 orders of magnitude!

Small fraction!

# Conduction in Intrinsic Silicon



- Conduction is due to **both electrons and holes** in a semiconductor

$$\sigma = qn\mu_n + qp\mu_p$$

Diagram illustrating the components of the conductivity equation:

- Electron density ( $n$ ) and Electron mobility ( $\mu_n$ ) contribute to the first term  $qn\mu_n$ .
- Hole density ( $p$ ) and Hole mobility ( $\mu_p$ ) contribute to the second term  $qp\mu_p$ .

Previously, we have only  $\sigma = qn\mu_n$

$\mu_n > \mu_p$

- Intrinsic silicon:  $\sigma = q(\mu_n + \mu_p)n_i$





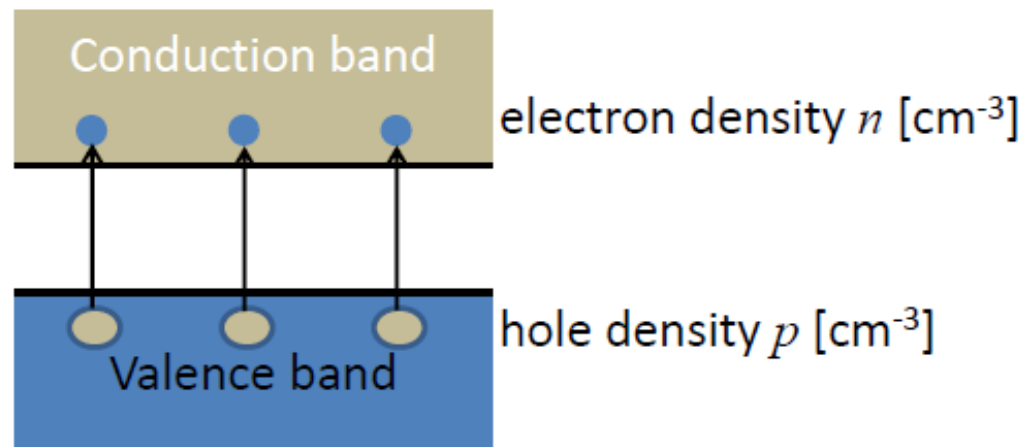
# Conduction in Intrinsic Silicon

$$J_n = qn\mu_n E$$

$$J_p = qp\mu_p E$$

$$J = J_n + J_p$$

$$J = \sigma E$$



$$\sigma = q(\mu_n n + \mu_p p)$$

Intrinsic Si@RT:  $\sigma = q(\mu_n + \mu_p)n_i$

$$= (1.6 \times 10^{19} \text{C}) \left( [1000 + 400] \frac{\text{cm}^2}{\text{Vs}} \right) (7 \times 10^9 \text{cm}^{-3})$$
$$= 1.5 \times 10^{-6} \text{ S/cm}$$

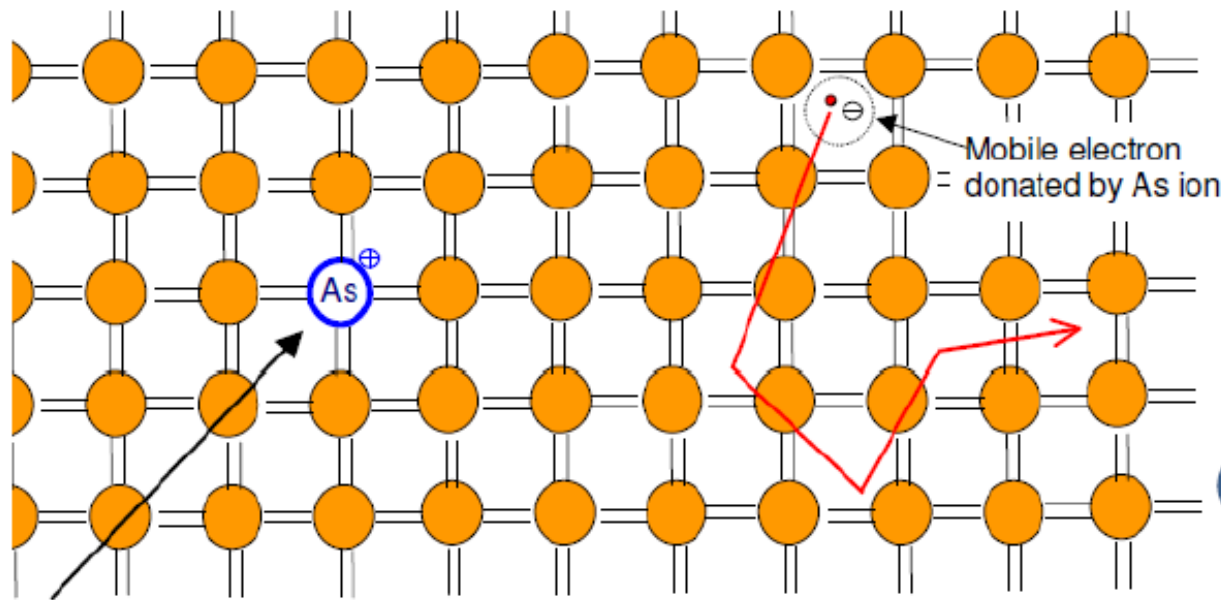
# Extrinsic Semiconductor

---

- Conductivity in intrinsic silicon is low in normal conditions (insulator).
  - Trace amounts of impurities (**dopants**) are added to adjust the conductivity.
  - Two types of substitutional dopants:
    - Group V elements, P, As, Sb (**donor**)
    - Group III elements, B, Al, Ga (**acceptor**)
-

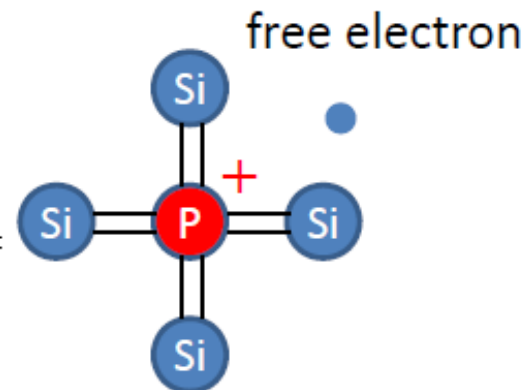
# Extrinsic Semiconductor - Donor

- Group V elements (**donors**)
  - Have one more outer electrons than Si (5 v.s. 4)
  - Can **donate** one free electron, and become positively charged (charge neutrality condition).



Immobile (stuck) positively charged arsenic ion after 5<sup>th</sup> electron left

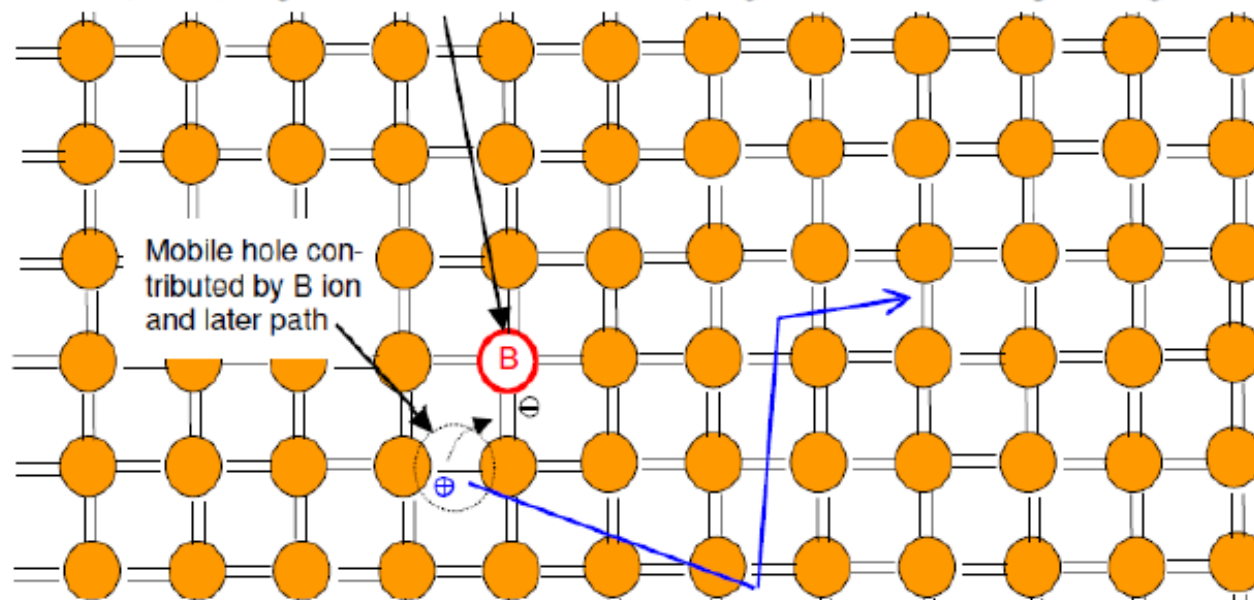
$N_D$  = donor density  
[cm<sup>-3</sup>] (**positive**,  
**fixed** charge)



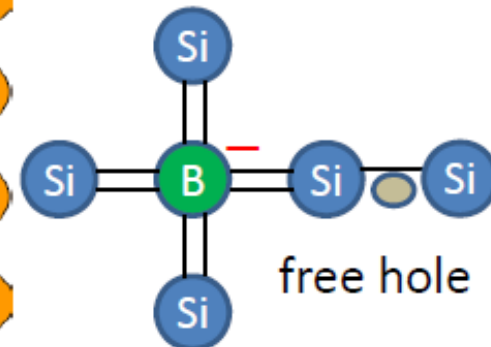
# Extrinsic Semiconductor - Acceptor

- Group III elements (**acceptors**)
  - Have one less outer electrons than Si (3 v.s. 4)
  - Can **accept** one electron (donate a hole), and become negatively charged (charge neutrality condition).

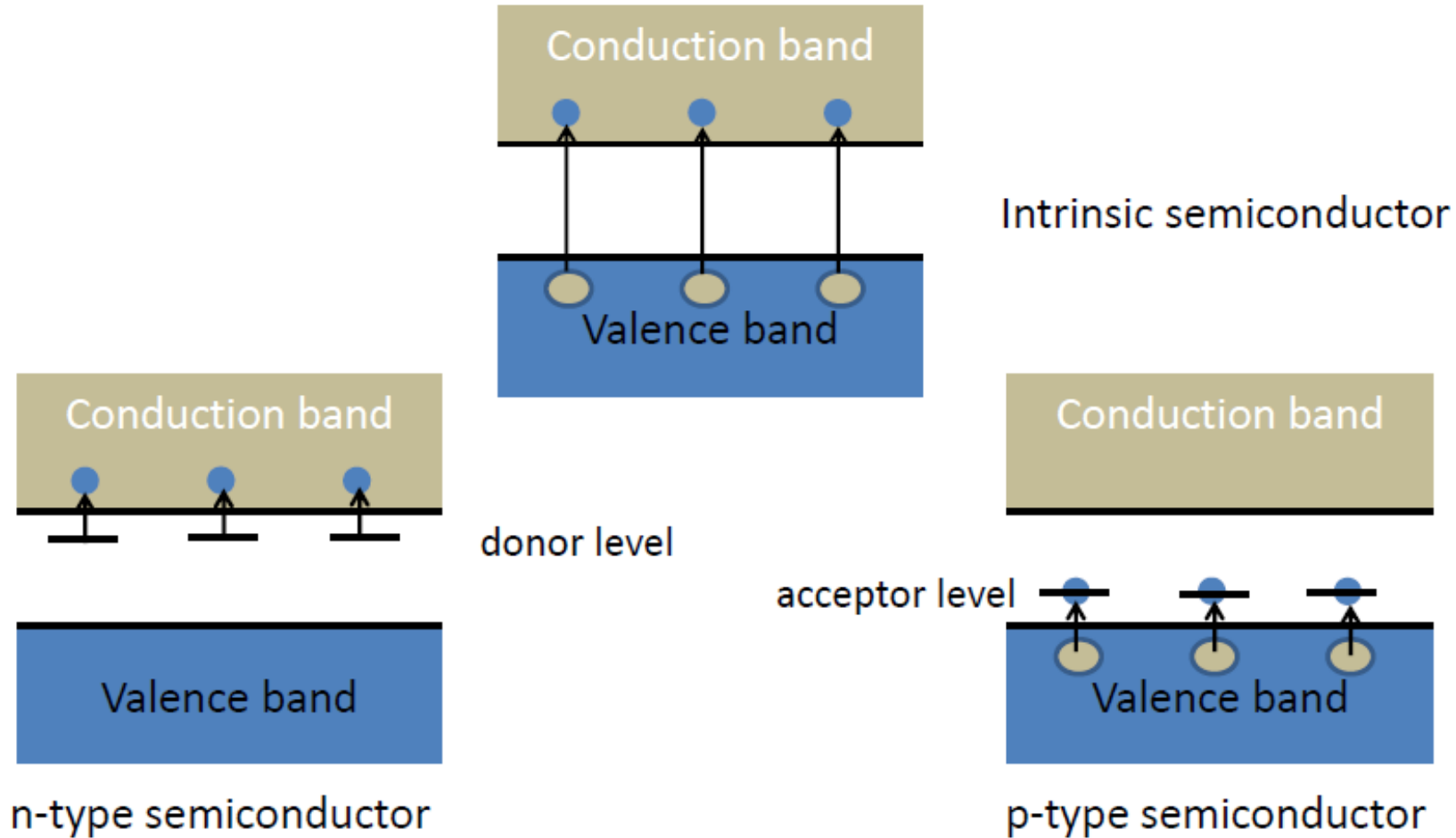
Immobile (stuck) negative boron ion after accepting electron from neighboring bond



$N_A$  = acceptor density [ $\text{cm}^{-3}$ ]  
(**negative**, fixed charge)

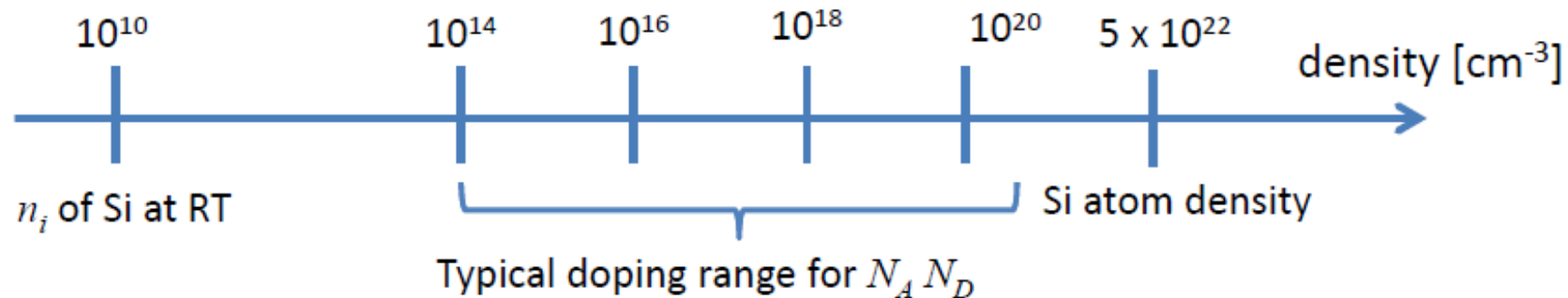


# Conduction in Intrinsic Silicon



# Typical Doping Range

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- Typical doping range:
    - Minimal chemical / mechanical changes (trace impurity)
    - Major changes in electrical conductivity
-

# Extrinsic Semiconductors

## – Carrier Density

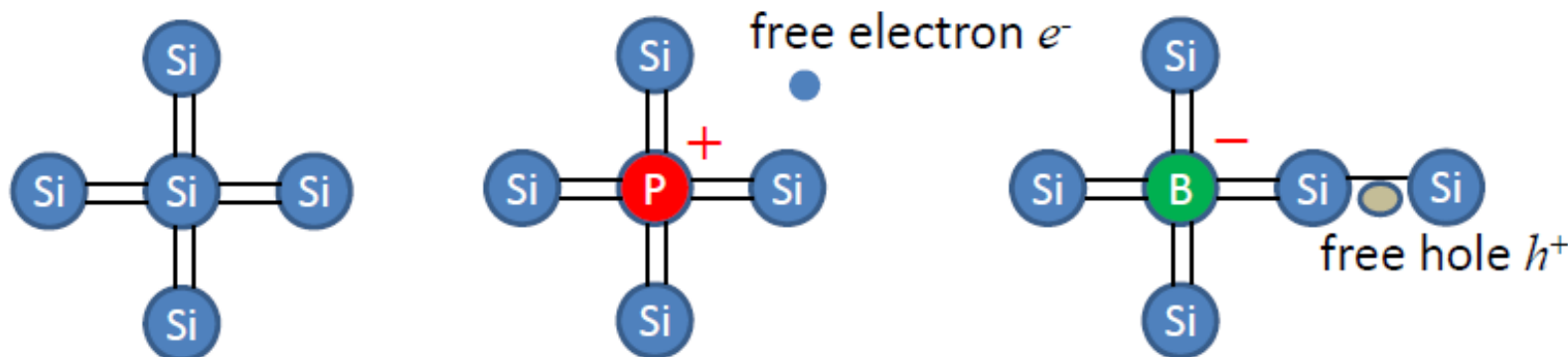
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- The carrier density ( $n, p$ ) in extrinsic semiconductors is governed by two laws:
  - Law of Mass Action (equilibrium condition):

$$np = n_i^2$$

- Charge Neutrality Condition:

$$N_D + p - N_A - n = 0$$





## Extrinsic Semiconductors – Carrier Density (*n*-type)

- Silicon doped with donors with density  $N_D$  ( $N_A = 0$ )
  - Solve  $np = n_i^2$  and  $N_D + p - n = 0$

– Eliminate  $p$ :

$$N_D + \frac{n_i^2}{n} - n = 0 \Rightarrow n^2 - nN_D - n_i^2 = 0$$

– Solve:  $n = \frac{N_D + \sqrt{N_D^2 + 4n_i^2}}{2}, p = \frac{n_i^2}{n}$  Negative solution tossed out

- Most of the time:  $N_D \gg n_i$ , thus  $n = N_D, p = \frac{n_i^2}{N_D} \Rightarrow n \gg n_i \gg p$ , the semiconductor is called “*n*-type”
- Electrons: **majority** carrier; Holes: **minority** carrier

## Extrinsic Semiconductors – Carrier Density (*p*-type)

- Silicon doped with acceptors with density  $N_A$  ( $N_D = 0$ )
  - Solve  $np = n_i^2$  and  $p - N_A - n = 0$

– Eliminate  $n$ :

$$p - N_A - \frac{n_i^2}{p} = 0 \Rightarrow p^2 - pN_A - n_i^2 = 0$$

– Solve:  $p = \frac{N_A + \sqrt{N_A^2 + 4n_i^2}}{2}, n = \frac{n_i^2}{p}$  Negative solution tossed out

- Most of the time:  $N_A \gg n_i$ , thus  $p = N_A, n = \frac{n_i^2}{N_A} \Rightarrow p \gg n_i \gg n$ , the semiconductor is called “*p*-type”
- Holes: **majority** carrier; Electrons: **minority** carrier

# Extrinsic Semiconductors

## – Carrier Density (compensation)

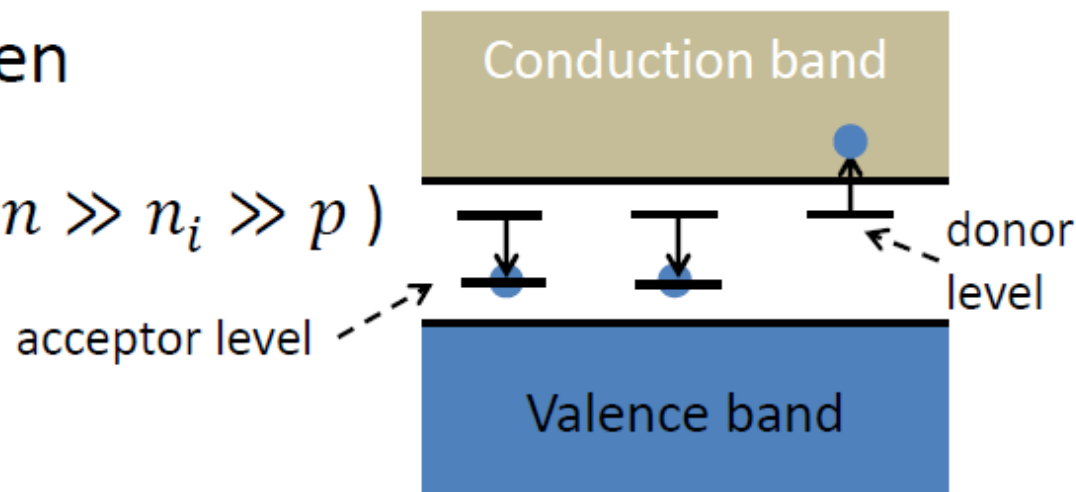
- Silicon doped with both donors ( $N_D$ ) and acceptors ( $N_A$ )
  - Suppose  $N_D > N_A$  (donors dominate)

$$n = \frac{(N_D - N_A) + \sqrt{(N_D - N_A)^2 + 4n_i^2}}{2} > n_i, \quad p = \frac{n_i^2}{n} < n_i \quad (\text{n-type})$$

– If :  $N_D - N_A \gg n_i$ , then

$$n = N_D - N_A, \quad p = \frac{n_i^2}{N_D - N_A} \quad (n \gg n_i \gg p)$$

- Compensation



Compensated semiconductor

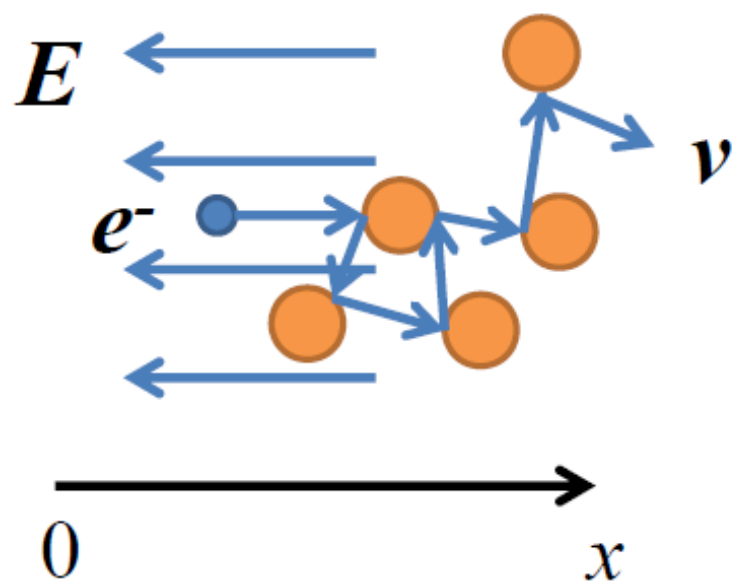
# Extrinsic Semiconductors – Carrier Density

---

- If the (net) dopant density is much larger than  $n_i$ , then the **majority** carrier density is equal to the dopant density.
  - The **minority** carrier density can then be obtained by the mass reaction law  $np = n_i^2$ .
-

# Recall: Electrons in Motion

## Constant electric field in a solid



Electron flies for short intervals ( $\sim$ ps) before bumping into **scattering objects**

Result: Electron's average velocity proportional to field

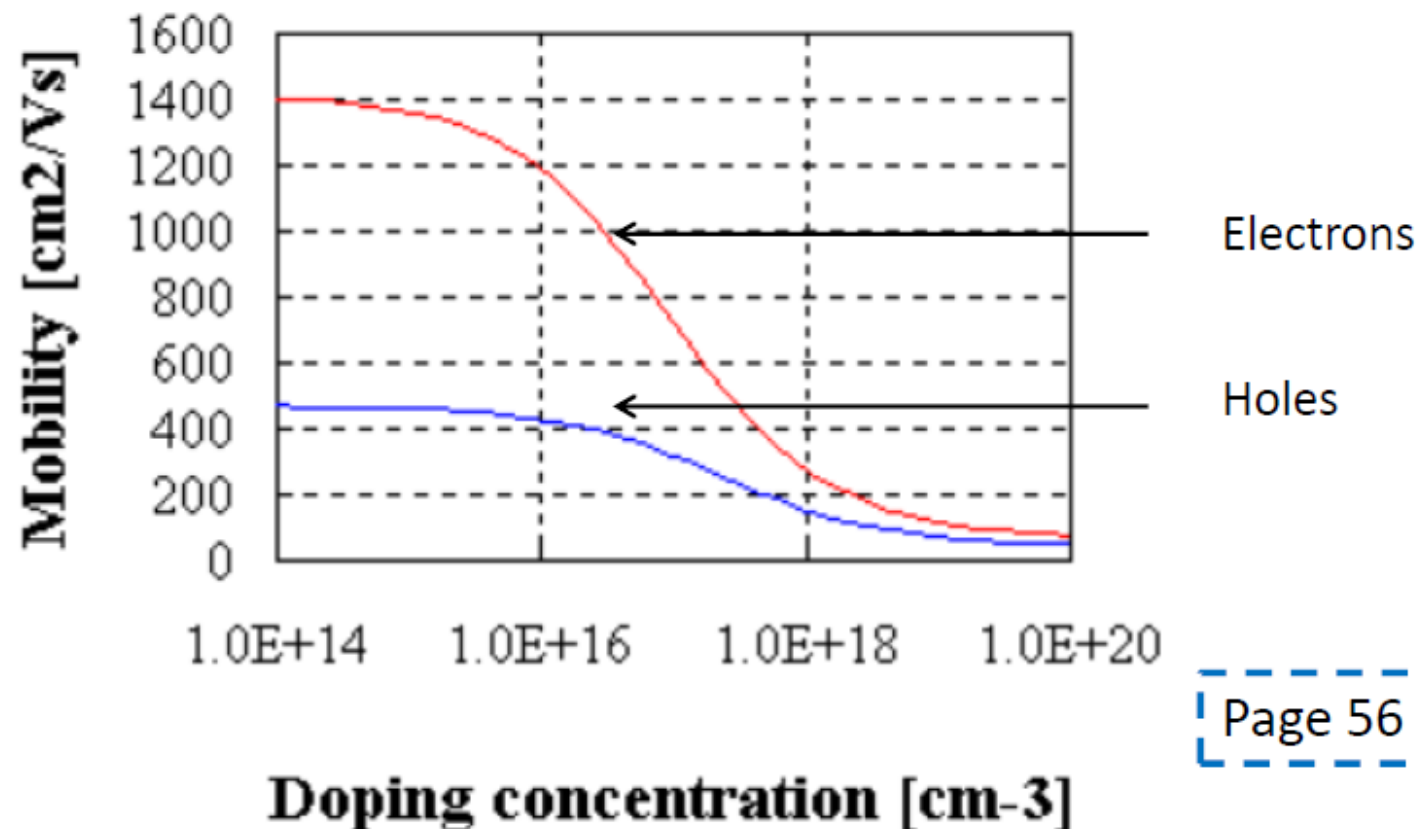
$$v = \mu E$$

# Extrinsic Semiconductors

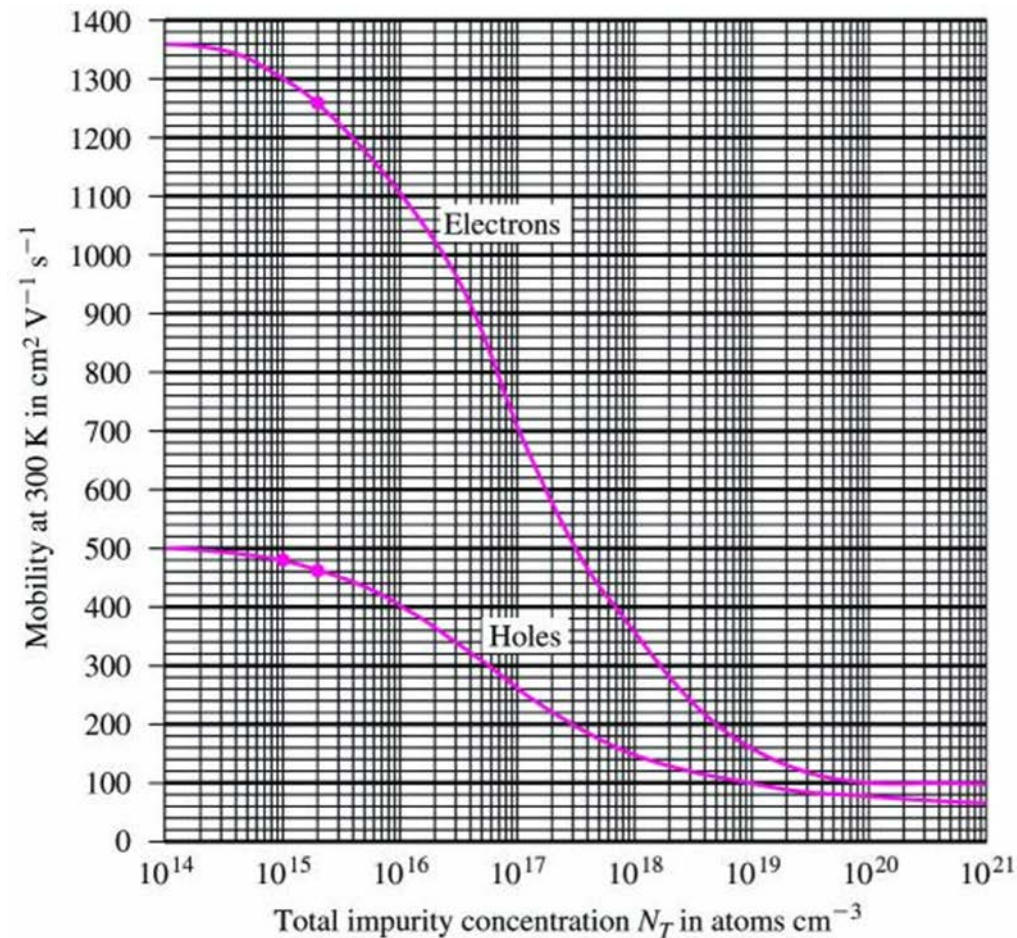
## Mobility & Resistivity

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- Impurities introduce scattering centers that degrade the mobility of **both** carriers ( $e^-$  &  $h^+$ ).



# Mobility and Resistivity in Doped Semiconductors



Mobility approximations

$$\mu_n = 92 + \frac{1270}{1 + \left( \frac{N_T}{1.3 \times 10^{17}} \right)^{0.91}}$$

$$\mu_p = 48 + \frac{447}{1 + \left( \frac{N_T}{6.3 \times 10^{16}} \right)^{0.76}}$$



# Extrinsic Semiconductors

## Mobility & Resistivity

---

- Doping affects resistivity via both mobility and density of carriers:

$$\sigma = q(n\mu_n + p\mu_p)$$

- Example: Si doped with As ( $N_D = 10^{17} \text{ cm}^{-3}$ ), what is the conductivity at 300 K?

**Step 1:** find  $n$  and  $p$  :

At 300 K,  $n_i = 1.0 \times 10^{10} \text{ cm}^{-3} \ll N_D$ , so

$$n = N_D = 10^{17} \text{ cm}^{-3}$$

$$p = n_i^2 / N_D = 1.0 \times 10^3 \text{ cm}^{-3}$$

---



# Extrinsic Semiconductors

## Mobility & Resistivity

---

**Step 2:** find  $\mu_n$  and  $\mu_p$ :

From mobility graph, find for  $N_D = 10^{17} \text{ cm}^{-3}$ :

$$\mu_n = 700 \text{ cm}^2/(\text{V}\cdot\text{s}); \quad \mu_p = 260 \text{ cm}^2/(\text{V}\cdot\text{s})$$

**Step 3:** Find  $\sigma$ :

$$\sigma = q(\mu_n n + \mu_p p)$$

$$\sigma = 1.6 \times 10^{-19} \text{ C} (10^{17} \text{ cm}^{-3} \times 700 \text{ cm}^2/\text{V}\cdot\text{s} + 1.0 \times 10^3 \text{ cm}^{-3} \times 260 \text{ cm}^2/\text{V}\cdot\text{s})$$

$$= 1.6 \times 10^{-19} \text{ C} (7 \times 10^{19} \text{ cm}^{-3} + \cancel{2.6 \times 10^5}) \text{ cm}^2/\text{V}\cdot\text{s}$$

$$= 11.2 \text{ } \Omega^{-1} \text{ cm}^{-1} \text{ (S/cm, } \mathfrak{U}/\text{cm)}$$

$$\rho = 1 / \sigma = 0.089 \text{ } \Omega \cdot \text{cm}$$

Hole contribution is negligible  
Electron contribution dominates

# Conduction Mechanisms

---

- **Drift** Current:

Electrons:  $j_n^{\text{drift}} = -qn v_n = -qn(-\mu_n E) = qn\mu_n E$

Holes:  $j_p^{\text{drift}} = qp v_p = qp(\mu_p E) = qp\mu_p E$

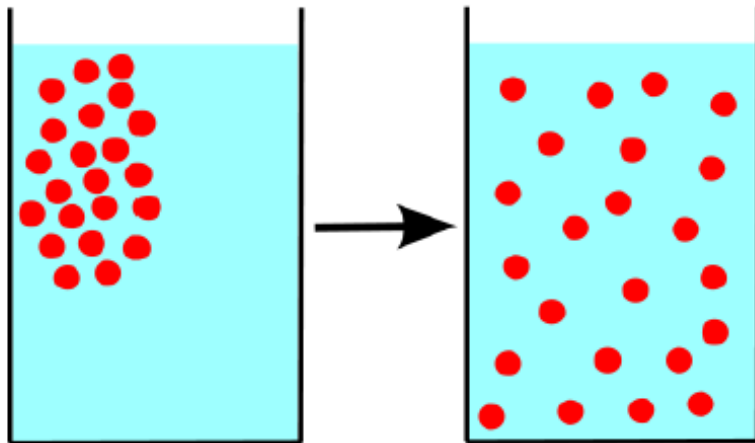
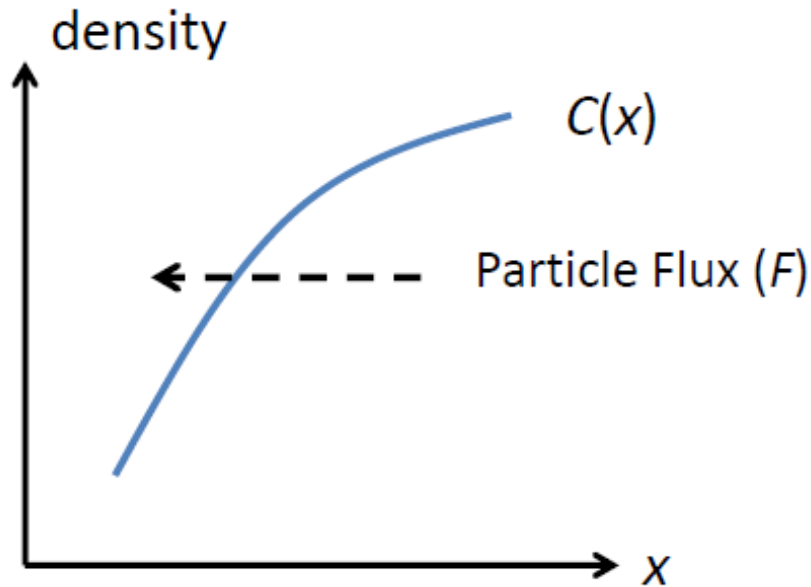
Total:  $j^{\text{drift}} = j_n^{\text{drift}} + j_p^{\text{drift}} = q \underbrace{(n\mu_n + p\mu_p)}_{\sigma} E$

Up to now we assumed that  $n, p$  are constant over space

When a spatial gradients of  $n$  and/or  $p$  exists, **diffusion** currents arise

---

# Diffusion



- Fick's Law of diffusion:

$$F = -D \frac{dC}{dx}$$

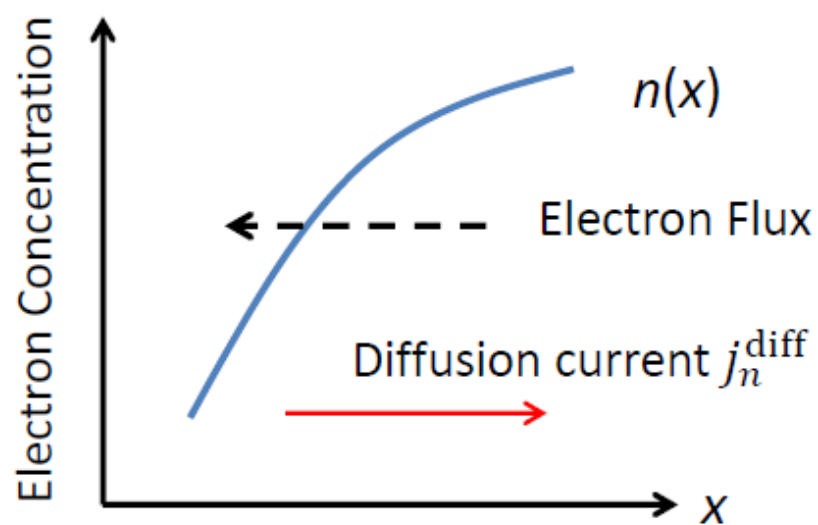
$F$ : Particle flux (amount per unit area per unit time) [ $\text{cm}^{-2}/\text{s}$ ]

$D$ : Diffusion coefficient or diffusivity [ $\text{cm}^2/\text{s}$ ]

$C$ : Concentration (density) of particles [ $\text{cm}^{-3}$ ]

Flux direction **opposite** to gradient

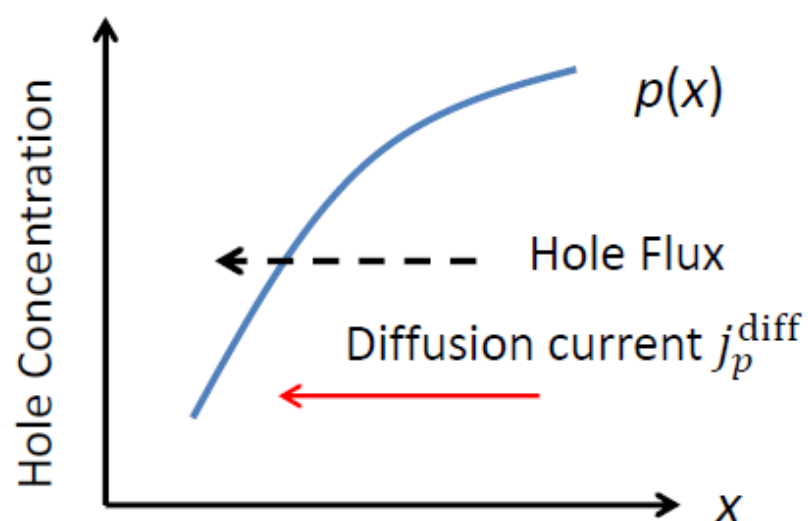
# Conduction Mechanisms – Diffusion Current



Electron flux:  $F_n = -D_n \frac{dn}{dx}$

Electron diffusion current:

$$j_n^{\text{diff}} = (-q)F_n = qD_n \frac{dn}{dx}$$



Hole flux:  $F_p = -D_p \frac{dp}{dx}$

Hole diffusion current:

$$j_p^{\text{diff}} = (+q)F_p = -qD_p \frac{dp}{dx}$$

Notice the sign difference!

# Charge Transport in Semiconductors

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- Total current = drift + diffusion

$$j_n^{\text{tot}} = j_n^{\text{drift}} + j_n^{\text{diff}} = qn\mu_n E + qD_n \frac{dn}{dx}$$

$$j_p^{\text{tot}} = j_p^{\text{drift}} + j_p^{\text{diff}} = qp\mu_p E - qD_p \frac{dp}{dx}$$

- 3-dimensional case:

$$\mathbf{j}_n^{\text{tot}} = qn\mu_n \mathbf{E} + qD_n \nabla n$$

$$\mathbf{j}_p^{\text{tot}} = qp\mu_p \mathbf{E} - qD_p \nabla p$$

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# Einstein Relationship

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- $D_n, D_p$ : diffusivity of electrons and holes
- $\mu_n, \mu_p$ : mobility of electrons and holes
- Einstein's relationship:

$$\frac{D}{\mu} = \frac{kT}{q}$$

Rhymes both forwards and backwards!

- $kT/q$  is called “thermal voltage” ( $V_T$ )
  - At 300 K,  $\frac{kT}{q} = \frac{8.62 \times 10^{-5} \text{ eV/K} \times 300 \text{ K}}{1e} = 25.9 \text{ mV}$
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